

## 用户提问



A note on commutative algebras and their modules in quasicategories.pdf

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how to understand?

## DEEPSEEK 回答

This paper by Saul Glasman presents a streamlined, combinatorial way to handle **modules over commutative algebras** in the setting of  $\infty$ -categories (specifically quasicategories). The main challenge it addresses is that while Lurie's *Higher Algebra* develops a good theory of modules over associative algebras using operads like  $\mathbf{LM}^{\otimes}$  and  $\mathbf{BM}^{\otimes}$ , the commutative case is more subtle. Glasman introduces a new operad  $\mathbf{CM}^{\otimes}$  that plays the analogous role for commutative algebras. Here's a conceptual breakdown to help you understand the key ideas, the main theorem, and the structure of the proof.

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### The Core Problem (in plain terms)

In ordinary category theory:

- A **commutative algebra**  $E$  in a symmetric monoidal category  $(\mathbf{C}, \otimes)$  is just an object with a multiplication  $E \otimes E \rightarrow E$  that is commutative and associative.
- An  $E$ -**module**  $M$  is an object with an action  $E \otimes M \rightarrow M$  that is compatible with the multiplication on  $E$ .
- Given such a pair  $(E, M)$ , you can define a functor from the category of finite pointed sets  $\mathbf{F}_*$  to  $\mathbf{C}$  by sending a pointed set  $S$  (with basepoint  $*$ ) to  $E^{\otimes(S^\circ)} \otimes M$ , where  $S^\circ = S \setminus \{*\}$ . This is a familiar construction in algebra (e.g.,  $E^{\otimes n} \otimes M$ ).

In  $\infty$ -categories, this is more subtle. Lurie already gave a definition of modules over commutative algebras using an operad called  $\mathbf{Mod}^{\mathbf{F}_*}(\mathbf{C})^{\otimes}$ , but Glasman wants a more explicit and functorial description that directly recovers the functor  $A_{E,M}(S) \simeq E^{\otimes S^\circ} \otimes M$ . Theorem 1 states that such a functor exists and is functorial.

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### The Key Idea: A New Operad $\mathbf{CM}^{\otimes}$

To encode the data of a commutative algebra  $E$  and a module  $M$  over it, Glasman defines a 1-category (then takes its nerve) called  $\mathbf{CM}^{\otimes}$ :

- **Objects:** Pairs  $(S, U)$  where  $S$  is a finite pointed set (with basepoint  $*$ ) and  $U \subseteq S^\circ$  is a subset of the non-basepoint elements.
  - Think of  $U$  as marking the positions where the module  $M$  sits; the rest of  $S^\circ$  are positions for copies of  $E$ .
  - Special cases:
    - $U = \emptyset$ : corresponds to having only  $E$ 's.
    - $|U| = 1$ : corresponds to exactly one  $M$  (and the rest  $E$ 's).
    - $|U| > 1$ : not allowed in the basic "pair" – but appears in the categorical structure.
- **Morphisms:** A map  $f: S \rightarrow T$  in  $\mathbf{F}_*$  (basepoint-preserving) such that **for each  $v \in V$  (the marked points in the target), the preimage  $U \cap f^{-1}(v)$  has exactly one element**. This condition ensures that the module factor  $M$  is not duplicated or lost – it propagates in a controlled way.

There is a natural forgetful functor  $CM^{\otimes} \rightarrow F_*$  sending  $(S, U)$  to  $S$ , making  $CM^{\otimes}$  an  $\infty$ -operad.

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### The Main Technical Step: Proposition 3

Proposition 3 is the heart of the paper. It says that the inclusion of a simpler suboperad  $(F_*)_{(1)^\circ}$  (the "slices" over the 1-point set) into  $CM^{\otimes}$  is a **trivial cofibration** in the model category of  $\infty$ -preoperads.

What does this mean?

- $(F_*)_{(1)^\circ}$  is the category of pointed finite sets  $S$  equipped with a map  $(1) \rightarrow S$  from the set with one non-basepoint. This effectively marks **at most one** element of  $S^\circ$  (or none if the map sends the unique element to the basepoint). So it corresponds to the special objects where  $|U| \leq 1$ .
- A trivial cofibration in this context means that **any map from  $(F_*)_{(1)^\circ}$  to an  $\infty$ -operad  $O^{\otimes}$  can be extended uniquely (up to homotopy) to a map from  $CM^{\otimes}$  to  $O^{\otimes}$** . In other words, the data of a commutative algebra and a module – which is exactly a map from  $(F_*)_{(1)^\circ}$  to  $C^{\otimes}$  – is equivalent to a map from  $CM^{\otimes}$  to  $C^{\otimes}$ .

This is what allows Glasman to prove Proposition 7:

$$\mathbf{Mod}(\mathbf{C}) \simeq \mathbf{Fun}^{\otimes}(CM^{\otimes}, C^{\otimes}).$$

That is, modules over a commutative algebra in  $\mathbf{C}$  are exactly  $CM^{\otimes}$ -algebras in  $\mathbf{C}$ .

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### Outline of the Proof of Proposition 3 (Lemmas 4–6 and Corollary)

The proof uses the language of **cocartesian fibrations** and **relative right Kan extensions**. Here's the intuition:

- **Lemma 4** is a lifting property for cocartesian fibrations: if you have a map from a cone  $K^{\infty}$  and you can lift the cone point and cocartesianly lift edges from the base, you can lift the whole diagram.
- **Lemma 5** shows that for any  $\infty$ -operad  $p: O^{\otimes} \rightarrow F_*$ , and any object  $X$  over  $S$ , the natural diagram  $\rho$  defined by taking cocartesian lifts along inert morphisms (like subset inclusions) is a  **$p$ -limit diagram** – it behaves like a "product diagram" in the operadic sense. This is proven by induction on the size of  $S^\circ$ , using the operad axioms for the base case  $|S^\circ| = 2$ .
- **Corollary 6** extends this: any nontrivial subcube (where the set difference has size  $> 1$ ) is also a  $p$ -limit diagram.

Now, to extend a map  $F$  from  $(F_*)_{(1)^\circ}$  to an operad  $O^{\otimes}$ , Glasman takes an arbitrary object  $(S, U)$  of  $CM^{\otimes}$  and constructs a certain poset  $Q_{(S,U)}$  of subsets of  $S^\circ$  that contains  $U$  and at most one extra element. This poset is coinital in the category of inert morphisms over  $(S, U)$ . Then he shows that the composition of  $F$  with this poset diagram embeds into a larger cube  $\rho$  on  $P_{S^\circ}$  (all subsets). By Corollary 6 and Lemma 5, the value at  $\rho_{S^\circ}$  (which is a product of copies of  $F(1, \{1\})$  and  $F(1, \emptyset)$ ) is a  $p$ -limit, hence the right Kan extension exists. This gives the extension.

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### The Symmetric Monoidal Envelope (Proposition 9, Corollary 10)

To get the explicit functor  $A_{E,M}: F_* \rightarrow \mathbf{C}$  with  $A_{E,M}(S) \simeq E^{\otimes S} \otimes M$ , Glasman computes the **symmetric monoidal envelope** of  $CM^{\otimes}$ . This is a construction that turns an operad into a symmetric monoidal category in a universal way.

He shows that the envelope is isomorphic to  $(F^+)^{\sqcup}$ , where:

- Objects of  $F^+$  are pairs  $(S, U)$  with  $S$  a finite set and  $U \subseteq S$  a subset.
- Morphisms  $(S, U) \rightarrow (T, V)$  are functions  $f: S \rightarrow T$  that restrict to a bijection  $U \cong V$ .
- The monoidal structure is disjoint union.

Then, by embedding  $F_*$  into  $F^+$  as those objects with  $|U| = 1$ , a  $\mathbf{CM}^\otimes$ -algebra (i.e., an  $(E, M)$  pair) gives a functor from  $F_*$  to  $\mathbf{C}$ . The formula  $E^{\otimes_S} \otimes M$  comes from interpreting the disjoint union in  $F^+$  as the tensor product in  $\mathbf{C}$ .

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## Summary: How to Read This Paper

1. **Understand the goal:** Theorem 1 – a clean, functorial description of modules over commutative algebras in  $\infty$ -categories.
2. **Learn the new operad  $\mathbf{CM}^\otimes$**  – objects are (pointed set  $S$ , subset  $U$  of non-basepoints). Morphisms preserve the marking condition.
3. **Proposition 3 is the key technical result** – the inclusion of the "at most one marked point" subcategory is a trivial cofibration, meaning any pair  $(E, M)$  extends uniquely to a  $\mathbf{CM}^\otimes$ -algebra.
4. **Proposition 7** connects this to Lurie's  $\mathbf{Mod}(\mathbf{C})$  – it's an equivalence of  $\infty$ -categories.
5. **Proposition 9 and Corollary 10** give the explicit tensor product formula by taking the symmetric monoidal envelope and restricting to the  $|U| = 1$  case.

The paper is dense but short. Focus on understanding the definitions of  $\mathbf{CM}^\otimes$  and the statement of Proposition 3. The lemmas (4–6) are technical but standard in Lurie's framework; they ensure that certain diagrams are  $p$ -limits, which is what allows the right Kan extension to work.

If you're comfortable with  $\infty$ -operads and cocartesian fibrations, this is a very neat example. If not, it might help to first study Lurie's *Higher Algebra*, especially §2.4, §3.3, and §4.2–4.3.

### 用户提问

how to understand marked morphism

### DEEPSEEK 回答

In the context of this paper (and Lurie's *Higher Algebra*), a **marked morphism** (or **marked edge**) is a technical tool used to define the model structure on  $\infty$ -**preoperads** and to handle cocartesian fibrations.

Let me break it down:

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## 1. Why “marked” edges appear

The category of  $\infty$ -preoperads (denoted  $\mathbf{PO}$ ) is a **marked simplicial set** model: objects are simplicial sets equipped with a distinguished set of edges called **marked edges**. This is similar to the model structure for marked simplicial sets used to encode  $\infty$ -categories with a specified class of cocartesian edges.

In Lurie's setup (see *Higher Algebra*, §2.1.4), an  $\infty$ -preoperad is a marked simplicial set  $p: \mathbf{O}^\otimes \rightarrow F_*$  satisfying certain properties, where the marked edges are precisely the **inert morphisms** (those that map to inert maps in  $F_*$ ). Marked edges play the role of “cocartesian edges over inert maps”. Thus, when Glasman says:

“We give  $(F_*)_{(1)/}$  the structure of an  $\infty$ -preoperad by letting the target map  $t: (F_*)_{(1)/} \rightarrow F_*$  **create marked edges**”

he means: **a morphism in  $(F_*)_{(1)/}$  is marked exactly when its image under  $t$  is an inert morphism in  $F_*$**  (i.e., a map of pointed finite sets that is a bijection on the non-basepoint elements, except possibly collapsing some to the basepoint). This is the standard way to inherit a marking from the target.

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## 2. Concrete example from the paper

Take  $(F_*)_{(1)/}$ . Its objects are maps  $j: \langle 1 \rangle \rightarrow S$  (where  $\langle 1 \rangle$  is the pointed set  $\{*, 1\}$ ). A morphism from  $j$  to  $k$  is a commutative triangle:

$$\begin{array}{ccc} \langle 1 \rangle & \xrightarrow{j} & S \\ & \searrow & \downarrow f \\ & \langle 1 \rangle & \xrightarrow{k} T \end{array}$$

So it's a map  $f: S \rightarrow T$  in  $F_*$  such that  $f \circ j = k$ .

Now,  $t$  sends this morphism to  $f$  in  $F_*$ . The morphism in  $(F_*)_{(1)/}$  is **marked** iff  $f$  is **inert** in  $F_*$ .

What does "inert" mean in  $F_*$ ?

A morphism  $f: S \rightarrow T$  is inert if it preserves the basepoint and for every  $t \in T^\circ$  (non-basepoint), the preimage  $f^{-1}(t)$  has exactly one element. In other words,  $f$  is a bijection from  $S^\circ$  onto  $T^\circ$  on the non-basepoint part (but can send some elements to the basepoint? Wait: careful – inert for  $F_*$  as an  $\infty$ -operad typically means:  $f$  is a bijection from  $S^\circ$  onto  $T^\circ$  – no extra basepoint-collapsing. Actually, in Lurie's definition, an inert morphism in  $F_*$  is a map that is a bijection on the non-basepoint elements and sends the basepoint to the basepoint. So it exactly restricts to an isomorphism  $S^\circ \cong T^\circ$ . That matches "inert" in the sense of operads: you don't multiply operations.)

Thus, in  $(F_*)_{(1)/}$ , an edge is marked exactly when the underlying map of pointed sets is a bijection on the non-basepoint parts.

## 3. Why is this useful?

Marked edges allow the use of **cocartesian liftings** in the model structure. In Lemma 4, Glasman refers to "cocartesian lift of  $e$  to  $\mathbf{E}$  with source  $I'$ ". For that to make sense,  $\mathbf{E} \rightarrow \mathbf{B}$  must be a **marked fibration** (or at least an inner fibration with a chosen class of marked edges that are cocartesian). By creating marked edges from the target, the map  $t: (F_*)_{(1)/} \rightarrow F_*$  becomes a **cocartesian fibration** (in the marked sense), which allows the lifting arguments in Lemma 4.

In short: **marked morphisms = the class of edges that are treated as "cocartesian by design"** in the  $\infty$ -preoperad structure. They are the ones you are allowed to lift uniquely (up to homotopy) when extending diagrams.

## 4. Intuition without technicalities

If you're not deep into model categories, think of it this way:

- In an  $\infty$ -operad, some morphisms are "inert" – they just rename positions or forget some, without any multiplication. These are the **marked** ones.
- When you build a new category like  $(F_*)_{(1)/}$  over  $F_*$ , you want to remember which maps are inert. So you "mark" exactly the ones that project to inert maps in the base.
- This marking is then used to define extensions (like the trivial cofibration in Proposition 3) and to guarantee that certain right Kan extensions exist.

So "create marked edges" simply means: *pull back the marking from the base category*.

## Summary

| Term               | Meaning in the paper                                                                                                         |
|--------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>Marked edge</b> | A distinguished 1-simplex that is to be treated as "cocartesian" or "inert" in the model structure for $\infty$ -preoperads. |

| Term                | Meaning in the paper                                                                                                     |
|---------------------|--------------------------------------------------------------------------------------------------------------------------|
| Create marked edges | The map $t$ defines a morphism in the source to be marked iff its image under $t$ is marked (i.e., inert) in the target. |

If you want to see the full definition of the model structure on marked simplicial sets, see Lurie’s *Higher Topos Theory*, §3.1, or *Higher Algebra*, §2.1.4. But for this paper, the key is: marking = inert morphisms, and “create” = pull back that class along  $t$ .

用户提问

how to understand all kinds of fibration?

DEEPSEEK 回答

To understand Glasman’s paper (and Lurie’s framework), you need a working knowledge of the main types of **fibrations** between simplicial sets (or marked simplicial sets). Each type encodes a different categorical or homotopical property. Here’s a concise guide.

## 1. Background: Simplicial Sets as $\infty$ -Categories

A **quasicategory** (or  $\infty$ -category) is a simplicial set where every inner horn ( $\Lambda^n_k$ ,  $0 < k < n$ ) has a filler. This replaces the usual composition law.

A **fibration** is a map of simplicial sets with a lifting property against certain horn inclusions. Different fibrations correspond to different lifting conditions.

## 2. The Main Fibrations in Order of Strength

| Fibration type        | Lifting against                                          | What it models                                                                                 |
|-----------------------|----------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Kan fibration         | all horns $\Lambda^n_k \rightarrow \Delta^n$             | Fibrations between Kan complexes ( $\infty$ -groupoids); “topological” fibrations.             |
| Left fibration        | $\Lambda^n_k \rightarrow \Delta^n$ for $0 \leq k < n$    | Categorical fibrations with contractible fibers; classify functors to $\infty$ -groupoids.     |
| Right fibration       | $\Lambda^n_k \rightarrow \Delta^n$ for $0 < k \leq n$    | Opposite of left fibrations.                                                                   |
| Inner fibration       | inner horns $\Lambda^n_k$ ( $0 < k < n$ )                | Allows fibers to be $\infty$ -categories, not just groupoids; used for cocartesian fibrations. |
| Cocartesian fibration | inner fibration + lifting of certain edges cocartesianly | Models “Grothendieck fibrations” in $\infty$ -categories; a functor to categories.             |
| Cartesian fibration   | dual to cocartesian.                                     |                                                                                                |
| Trivial fibration     | all inclusions $\partial\Delta^n \rightarrow \Delta^n$   | Weak equivalence and fibration; acyclic in model structure.                                    |

## 3. Detailed Descriptions

## Inner fibration – the foundation

A map  $p : X \rightarrow Y$  is an **inner fibration** if it has the right lifting property against the inner horn inclusions  $\Lambda_k^n \hookrightarrow \Delta^n$  for  $0 < k < n$ .

- **Purpose:** If  $Y$  is an  $\infty$ -category, then every fiber  $p^{-1}(y)$  is also an  $\infty$ -category.
- **Example:** Any map between quasicategories that is a categorical fibration is an inner fibration.
- **In Glasman:** The map  $\mathbf{CM}^{\otimes} \rightarrow \mathbf{F}_*$  is an inner fibration (since it's an  $\infty$ -operad). Also, Lemma 4 starts with an inner fibration  $q : \mathbf{E} \rightarrow \mathbf{B}$ .

## Cocartesian fibration – the key for indexed categories

A map  $p : X \rightarrow Y$  is a **cocartesian fibration** if:

1.  $p$  is an inner fibration.
  2. For every edge  $e : y_1 \rightarrow y_2$  in  $Y$  and every  $x_1 \in X$  with  $p(x_1) = y_1$ , there exists a **p-cocartesian edge**  $\tilde{e} : x_1 \rightarrow x_2$  lifting  $e$  (meaning that for any  $x_3$ , composing with  $\tilde{e}$  induces a homotopy equivalence of mapping spaces).
- **Intuition:** It's the  $\infty$ -categorical version of a Grothendieck fibration where fibers vary covariantly (hence "co").
  - **Example:** The projection  $(\mathbf{F}_*)_{(1)/} \rightarrow \mathbf{F}_*$  (with appropriate markings) is a cocartesian fibration.
  - **Used in:** Lemma 4, Lemma 5 (to lift diagrams along cocartesian edges).

## Trivial fibration (also called acyclic fibration)

A map that has the right lifting property against all boundary inclusions  $\partial\Delta^n \hookrightarrow \Delta^n$ .

- **Equivalently:** It is a Kan fibration and a weak homotopy equivalence.
- **In model categories:** It's a trivial cofibration in the dual sense.
- **In Glasman:** Proposition 3 shows  $\phi$  is a **trivial cofibration** in the PO model structure – that means the opposite map (if you think of the injective direction) is a trivial fibration in the dual. Actually, in Proposition 7,  $\phi^*$  is a **trivial Kan fibration** of simplicial sets (meaning it has the RLP against all  $\partial\Delta^n \hookrightarrow \Delta^n$ ). That's why it's an equivalence of mapping spaces.

## Kan fibration

Lifts against all horns  $\Lambda_k^n \hookrightarrow \Delta^n$  (including outer horns).

- **Fibers are Kan complexes** ( $\infty$ -groupoids).
- **Example:** Any map between Kan complexes that is a Serre fibration in topology.
- In Glasman's Proposition 7, the restriction map  $\phi^*$  is a **trivial Kan fibration** – thus a homotopy equivalence and a fibration, so it's an acyclic fibration in the Kan model structure.

## Marked fibrations (from *Higher Algebra*)

When we work with **marked simplicial sets** (simplicial sets with a distinguished set of edges), a marked fibration is a map that lifts marked anodyne extensions (certain inclusions involving marked edges).

- **Purpose:** To control cocartesian edges in a combinatorial way.
- In Glasman: The  $\infty$ -preoperad model structure is a marked model structure. The map  $\phi$  being a trivial cofibration in PO means it is a marked anodyne extension.

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## 4. How They Relate

A **trivial fibration** is the strongest: it's both a fibration and a weak equivalence.

A **cocartesian fibration** is an inner fibration with extra lifting of edges, but not necessarily a Kan fibration.

An **inner fibration** is the weakest of the bunch (still strong enough to guarantee  $\infty$ -category fibers).

In Lurie's theory, a **cocartesian fibration** is the right notion for "pseudofunctor to  $(\infty, 1)\text{-Cat}$ ".

Lemma 4 uses the fact that if you can lift edges cocartesianly, you can lift whole cones.

## 5. Practical Advice for Reading Glasman

| Where it appears | What fibration                                                                                                               | Why                                                                                              |
|------------------|------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Lemma 4          | Inner fibration $q : \mathbf{E} \rightarrow \mathbf{B}$ ; also cocartesian edges are assumed to exist.                       | To lift $K^\infty$ via cocartesian lifts.                                                        |
| Lemma 5          | $p : \mathcal{O}^\otimes \rightarrow \mathbf{F}_*$ is an $\infty$ -operad, hence an inner fibration with marked inert edges. | To talk about $p$ -limit diagrams (limits relative to $p$ ).                                     |
| Proposition 3    | $\phi$ is a trivial cofibration in $\mathbf{PO}$ (marked model structure).                                                   | So that every map from $(\mathbf{F}_*)_{(1)^\vee}$ extends uniquely over $\mathbf{CM}^\otimes$ . |
| Proposition 7    | $\phi^*$ is a trivial Kan fibration.                                                                                         | So restriction is an equivalence of $\infty$ -categories.                                        |

## 6. A Mnemonic

- **Inner fibration** = "category-like" fibers.
- **Cocartesian fibration** = "functor to  $\mathbf{Cat}$ " (covariant).
- **Cartesian fibration** = "functor to  $\mathbf{Cat}$ " (contravariant).
- **Kan fibration** = "groupoid-like" fibers.
- **Trivial fibration** = "homotopy equivalence that is also a fibration".

In the paper, you can usually ignore the fine details of model categories as long as you remember:

- **Trivial cofibration**  $\Rightarrow$  left homotopy equivalence that is a cofibration; its opposite (or the induced map on mapping spaces) is a trivial fibration.
- **Cocartesian lifts** allow you to spread a "product" structure along inert maps.

If you want to go deeper, read Lurie's *Higher Topos Theory* §2.1–2.4 (fibrations) and *Higher Algebra* §2.4 (cocartesian fibrations for operads). But for Glasman's note, the above distinctions should be enough.

### 用户提问

how to use these fibrations? what are they used for?

### DEEPSEEK 回答

These fibrations are the main technical machinery that makes **homotopy-coherent mathematics** work. Without them, you cannot:

- **Lift diagrams** along functors,



- **Define limits/colimits** parametrized by another category,
- **Prove extension theorems** (like Proposition 3),
- **Model indexed categories** (e.g., a family of objects varying over a base  $\infty$ -category).

Let's go through each fibration type and see **exactly what problem it solves**, then show how Glasman uses them.

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## 1. Inner fibration – the bare minimum to be an $\infty$ -category

### What is it for?

If a map  $p: X \rightarrow Y$  is an **inner fibration**, then:

- Each fiber  $p^{-1}(y)$  is an  $\infty$ -category (because inner horns can be filled).
- Moreover,  $X$  itself is an  $\infty$ -category if  $Y$  is.

### Role in practice:

Almost every structure in Higher Algebra is built over the base  $F_*$  (finite pointed sets) via an inner fibration. An  $\infty$ -operad is exactly a map  $O^{\otimes} \rightarrow F_*$  that is an inner fibration and satisfies certain “inert edge” conditions.

### In Glasman:

- $CM^{\otimes} \rightarrow F_*$  is an inner fibration (by definition of  $\infty$ -operad).
- Lemma 4 starts with an inner fibration  $q: E \rightarrow B$  because we need the fibers to be  $\infty$ -categories to talk about (co)cartesian edges.

**Takeaway:** Inner fibration = “the target is a good index category and the map respects composition enough that fibers are  $\infty$ -categories”.

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## 2. Cocartesian fibration – the mother of all “indexed categories”

### What is it for?

A cocartesian fibration  $p: X \rightarrow Y$  is the  $\infty$ -categorical version of a **Grothendieck fibration** (covariant version). It allows you to:

- **Push forward** objects along morphisms in  $Y$  (via cocartesian lifts).
- **Represent** a functor  $Y \rightarrow \mathbf{Cat}_{\infty}$  (the “Straightening–Unstraightening” correspondence).

### Concrete use in the paper:

In Lemma 4 and Lemma 5, Glasman wants to take a diagram in the base  $F_*$  (like the poset of subsets) and **lift it** to a diagram in  $O^{\otimes}$  (the operad) while maintaining control over edges. The key property: for each edge  $e$  in the base, there exists a **cocartesian lift** that is unique up to homotopy. This lets you “spread” an object  $X$  (like  $E^{\otimes U} \otimes M$ ) over all subsets by repeatedly taking cocartesian lifts.

### Example:

Given  $X \in O_S^{\otimes}$  and an inert map  $S \rightarrow T$  (e.g., including a subset), you get a canonical morphism  $X \rightarrow \text{some object over } T$  that is cocartesian. Repeating this for all inclusions builds the whole cube  $\rho$  in Lemma 5.

### Why cocartesian and not just cartesian?

Because the direction of the index category (subsets) goes **from smaller to larger** (or vice versa). In Lemma 5,  $r_S: P_{S^o} \rightarrow F_*$  sends a subset  $U$  to the pointed set where  $U$  are the active points. This map is **covariant** in  $U$  (larger subset gives a map “forgetting” the other points). Cocartesian fibrations push forward.

**Takeaway:** Cocartesian fibration = “a family of objects parametrized by  $Y$ , with a well-defined way to go forward along morphisms”.



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### 3. Trivial fibration (trivial Kan fibration) – encoding equivalences

#### What is it for?

A trivial fibration is a **weak equivalence that is also a fibration**. In model categories, trivial fibrations are the maps that lift against all cofibrations. In practice, if  $f: A \rightarrow B$  is a trivial fibration of simplicial sets, then:

- $f$  is a homotopy equivalence,
- $f$  has the right lifting property against every  $\partial\Delta^n \hookrightarrow \Delta^n$ ,
- So  $f$  is surjective on all homotopy groups and on all simplicial sets.

#### Concrete use in Glasman:

In Proposition 7,  $\phi^*: \text{Fun}^\otimes(\text{CM}^\otimes, \mathbf{C}^\otimes) \rightarrow \text{Fun}^\otimes((F_*)_{(1)/}, \mathbf{C}^\otimes)$  is a **trivial Kan fibration**. That means:

- It is an equivalence of  $\infty$ -categories (since trivial fibrations are categorical equivalences).
- It is surjective on objects up to equivalence.
- It has lifts for all horn inclusions, meaning any map from  $(F_*)_{(1)/}$  extends (up to homotopy) to a map from  $\text{CM}^\otimes$  **uniquely** (in the  $\infty$ -categorical sense).

Without the “trivial” part, you would only have a fibration that might not be an equivalence. The “trivial” guarantees that the restriction map is an **acyclic fibration** in the model category, hence a weak equivalence.

**Takeaway:** Trivial fibration = “this map is a homotopy equivalence and a fibration, so you can lift any diagram from the base to the total space” .

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### 4. Marked fibrations (in the $\infty$ -preoperad model structure)

#### What is it for?

Marked simplicial sets add a set of distinguished edges (the “inert” edges for operads). A **marked fibration** respects those markings: it lifts against marked anodyne extensions (inclusions that generate the marked model structure).

#### Why mark edges?

In an  $\infty$ -operad, not every map is created equal: **inert maps** (those that are bijections on the non-basepoint part) behave like “projections” or “inclusions of positions” . You want these to be the ones that are **cocartesian** (i.e., you can lift them uniquely). Marking them tells the model category which edges to treat as “good” .

#### Concrete use:

In Proposition 3,  $\phi$  is a **trivial cofibration** in the marked model structure of  $\infty$ -preoperads. That means:

- $\phi$  is a weak equivalence and a cofibration (in the marked sense).
- Therefore, **any map from  $(F_*)_{(1)/}$  to an  $\infty$ -operad (with its natural marking) can be extended to a map from  $\text{CM}^\otimes$**  because the extension problem becomes a lifting problem against a trivial cofibration.

**Takeaway:** Marked fibrations/cofibrations are the language that makes operadic extensions “just homotopy theory” .

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### Summary Table: What each fibration is used for in Glasman’s paper

| Fibration type               | Where used                              | Purpose                                                                                                       |
|------------------------------|-----------------------------------------|---------------------------------------------------------------------------------------------------------------|
| Inner fibration              | Lemma 4,<br>Lemma 5                     | Ensure base and fibers are $\infty$ -categories; allow definition of $p$ -limits.                             |
| Cocartesian fibration        | Lemma 4,<br>Lemma 5                     | Lift diagrams from the index category (e.g., subset poset) to the operad; propagate objects along inert maps. |
| Trivial (Kan) fibration      | Proposition 7                           | Prove that restriction along $\phi$ is an equivalence of $\infty$ -categories; guarantee unique extensions.   |
| Marked fibration/cofibration | Proposition 3,<br>PO model<br>structure | Formalize the “inert edges” as marked; allow $\phi$ to be a trivial cofibration so that extensions exist.     |

## A Concrete Analogy (for 1-categories)

In ordinary category theory:

- A **Grothendieck fibration**  $p: E \rightarrow B$  is the 1-categorical version of a cocartesian fibration. It lets you “lift” morphisms from  $B$  to  $E$  **cartesially** or **cocartesially**.
- A **trivial fibration** in the model structure on categories (e.g., an equivalence that is surjective on objects) would be something like a functor that is an isomorphism on the underlying nerve after taking Kan complexes.

In  $\infty$ -categories, you cannot directly say “take  $p^{-1}(y)$  as a category” because composition is not strictly associative. The fibration conditions encode **all the higher coherence**.

So when you see a fibration in Lurie or Glasman, think: **this map has a lifting property that allows me to move objects along edges in the base without breaking homotopy coherence**. And the specific type tells you **which direction** (cocartesian = forward) and **how strong** the lifting is (trivial = equivalence).

If you want to master these, practice with:

- *Higher Topos Theory* §2.4.2 (cocartesian fibrations)
- *Higher Algebra* §2.4 (operads as fibrations over  $F_*$ )
- The “straightening–unstraightening” equivalence (HTT §3.2).

### 用户提问

can I just viewed them as some “functors” for intuition?

### DEEPSEEK 回答

Yes, you can — and for intuition, you **should** view them as “functors with extra structure”. But the exact analogy depends on the fibration type.

Here’s a simple **intuition table** (1-category vs.  $\infty$ -category):

| 1-category notion                                              | $\infty$ -categorical analog (as a fibration)    | What it “acts like”                                                                                                                                                                                                                                                                                                   |
|----------------------------------------------------------------|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A functor $F : B \rightarrow \mathbf{Cat}$                     | Cocartesian fibration<br>$E \rightarrow B$       | A family of categories (fibers) that vary covariantly over $B$ .                                                                                                                                                                                                                                                      |
| A functor $F : B \rightarrow \mathbf{Cat}^{\mathrm{op}}$       | Cartesian fibration<br>$E \rightarrow B$         | Same, but contravariant.                                                                                                                                                                                                                                                                                              |
| An ordinary functor $F : B \rightarrow C$ (no parametrization) | Inner fibration with discrete fibers? Not quite. | Actually: any functor between $\infty$ -categories is an <b>inner fibration</b> after passing to the twisted arrow category? No – better: A functor $F : C \rightarrow D$ of $\infty$ -categories is an inner fibration of simplicial sets (because any functor between quasi-categories is an inner fibration). Yes! |
| A natural isomorphism                                          | Trivial fibration                                | A “homotopy equivalence” that is also a fibration – think of a functor that is an equivalence and surjective on objects “in a nice way”.                                                                                                                                                                              |
| A map of operads (preserving inert maps)                       | Marked fibration / trivial cofibration in PO     | A “functor that respects the special inert edges”.                                                                                                                                                                                                                                                                    |

## Breaking it down intuitively

### 1. Cocartesian fibration = “a functor to $\mathbf{Cat}_{\infty}$ ”

In 1-category theory, a **Grothendieck fibration**  $p : E \rightarrow B$  corresponds to a functor  $B \rightarrow \mathbf{Cat}$ .

The  $\infty$ -version: a **cocartesian fibration**  $p : E \rightarrow B$  corresponds to a functor  $B \rightarrow \mathbf{Cat}_\infty$  (the  $\infty$ -category of  $\infty$ -categories).

So when you see a cocartesian fibration, think: “This is the total space of a family of  $\infty$ -categories indexed by  $B$ , with pushforwards along morphisms.”

**In Glasman:**

The projection  $(F_*)_{(1)/} \rightarrow F_*$  (with marking created from inert maps) is a cocartesian fibration.

Intuitively, it’s the family that sends each pointed set  $S$  to the category of “ways to mark one element or none”.

## 2. Inner fibration = “a functor between $\infty$ -categories (or an $\infty$ -operad)”

Any functor  $F : C \rightarrow D$  between quasi-categories is an **inner fibration** (Lurie, HTT). So inner fibration is the minimal structure to say “this map is a functor of  $\infty$ -categories”.

In the context of operads: the structure map  $O^\otimes \rightarrow F_*$  is an inner fibration – so you can just think of it as “the operad’s underlying functor to the category of finite pointed sets”.

## 3. Trivial fibration = “an equivalence that is well-behaved”

A trivial fibration  $X \rightarrow Y$  (e.g., a trivial Kan fibration) is a **weak equivalence** and a **fibration** – think: a map that looks like a projection from a product of a contractible space onto one factor, or a retraction of a deformation. In practice, it means “you can lift any diagram from  $Y$  to  $X$ , and the lift is essentially unique”.

So when you see it in Proposition 7:  $\phi^*$  is a trivial Kan fibration – that means **restriction along  $\phi$  is an equivalence of mapping spaces**; it’s like a “bijection on homotopy classes of maps”.

## 4. Marked fibration / trivial cofibration = “a map of operads that respects the special edges”

Marked edges are the “inert” morphisms (those that behave like inclusions or projections). A marked fibration is a map that lifts against **marked anodyne extensions** – intuitively, it’s a functor that preserves the property of being “inert” in a homotopy-coherent way.

A **trivial cofibration** in the marked model structure is an “anodyne extension” – it’s like attaching cells that are homotopically harmless. So when Glasman shows  $\phi$  is a trivial cofibration, he’s saying: “ $(F_*)_{(1)/}$  sits inside  $CM^\otimes$  as a kind of ‘skeleton’, and adding the rest of  $CM^\otimes$  does not change the homotopy theory of maps out of it.”

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## So, can you just view them as “functors” for intuition?

Yes, with a mental footnote:

- **Cocartesian fibration**  $\rightarrow$  “functor to  $\mathbf{Cat}_\infty$ ” (covariant family).
- **Cartesian fibration**  $\rightarrow$  “functor to  $\mathbf{Cat}_\infty^{\text{op}}$ ” (contravariant family).
- **Inner fibration**  $\rightarrow$  “functor between  $\infty$ -categories” (or an  $\infty$ -operad over  $F_*$ ).
- **Trivial fibration**  $\rightarrow$  “equivalence that is a fibration” (think: projection from a product of a contractible space).
- **Marked fibration / trivial cofibration**  $\rightarrow$  “map of operads that respects inert edges, up to homotopy”.

In everyday reasoning, you can **replace “cocartesian fibration  $E \rightarrow B$ ” with “a functor  $B \rightarrow \mathbf{Cat}_\infty$ ”** without losing much intuition – the technical details ensure that you can pull back, push forward, and compose coherently.

Just remember: in  $\infty$ -categories, “functor” itself is defined via inner fibrations (or via simplicial mapping spaces), so the fibration language is actually the **foundation**, not just an analogy. But for

intuition, the functor viewpoint is perfect.

#### 用户提问

the cases to use different fibrations? what is the principal?

#### DEEPSEEK 回答

Excellent question. The **principle** is:

**Choose the fibration type based on what structure you need to lift and what kind of diagrams you want to represent.**

In short:

- **Inner fibration** → I just need the target to be an  $\infty$ -category and fibers to be  $\infty$ -categories.
- **Cocartesian (or Cartesian) fibration** → I need to model a family of objects that varies **functorially** along morphisms in the base.
- **Trivial fibration** → I need a **weak equivalence** that is also a fibration (for lifting all diagrams uniquely up to homotopy).
- **Marked fibration** → I am working with **operads** or other structures where certain edges are “inert” and must be preserved.

Let’s go case by case.

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## 1. Inner fibration – the default for “functor between $\infty$ -categories”

**Use when:** You only need that the map behaves like a functor and that fibers are  $\infty$ -categories. No requirement to lift special classes of edges uniquely.

**Examples in the paper:**

- $p: \mathcal{O}^{\otimes} \rightarrow \mathcal{F}_*$  for an  $\infty$ -operad.
- The map in Lemma 4 is an inner fibration so that the fibers are  $\infty$ -categories and cocartesian edges can be defined relative to it.

**Principle:** “I want to talk about objects over a base that themselves form  $\infty$ -categories, but I don’t need a Grothendieck construction yet.”

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## 2. Cocartesian fibration – for “indexed $(\infty, 1)$ -categories”

**Use when:** You need to **push forward** objects along morphisms in the base, and you want that assignment to be **functorial up to coherent homotopy**. That is, you are doing the  $\infty$ -version of a Grothendieck fibration.

**Examples in the paper:**

- $(\mathcal{F}_*)_{(1)/} \rightarrow \mathcal{F}_*$  (with marking created from inert maps) is a cocartesian fibration.
- Lemma 4 and Lemma 5 rely heavily on cocartesian lifts to propagate an object  $X$  over  $S$  to all subsets  $U \subseteq S^{\circ}$ .

**Principle:** “I need to associate to each object  $b$  in the base a fiber  $X_b$ , and to each morphism  $b \rightarrow b'$  a transfer functor  $X_b \rightarrow X_{b'}$  (covariantly).” This is exactly what a cocartesian fibration encodes (via straightening-unstraightening).

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## 3. Cartesian fibration – dual, for contravariant indexing

**Use when:** You need to **pull back** objects along morphisms (contravariant transfer). This is just the opposite of a cocartesian fibration.

**In the paper:** Not directly used, but Lurie's operad theory also uses cartesian fibrations for right modules. The principle is symmetric.

**Principle:** "I want a contravariant functor  $B^{\text{op}} \rightarrow \mathbf{Cat}_\infty$ ."

---

## 4. Trivial fibration – for “equivalence with lifting property”

**Use when:** You want a map that is a **weak equivalence** and also a fibration, so that it lifts **all** diagrams from the base (including boundaries) uniquely up to homotopy. This appears in **extension theorems** and **equivalence of  $\infty$ -categories**.

**Examples in the paper:**

- $\phi^*$  in Proposition 7 is a **trivial Kan fibration**. This proves that restriction along  $\phi$  is an equivalence of mapping spaces (hence an equivalence of  $\infty$ -categories).
- In model category language, trivial fibrations are the **acyclic fibrations** – they are the “nice surjective equivalences”.

**Principle:** “I want to show that two  $\infty$ -categories (or spaces of maps) are equivalent, and I can do so by exhibiting a fibration that is also a weak equivalence.”

---

## 5. Marked fibration / trivial cofibration – for operadic contexts

**Use when:** You are working in the  **$\infty$ -preoperad model structure** (or any marked simplicial set model) where certain edges are designated as “inert” or “cocartesian”. Marked fibrations lift against **marked anodyne extensions**, which respect these distinguished edges.

**Examples in the paper:**

- The map  $\phi : (F_*)_{(1)/} \rightarrow \mathbf{CM}^\otimes$  is a **trivial cofibration** in the marked model structure PO. This allows any map from  $(F_*)_{(1)/}$  to an  $\infty$ -operad to extend to a map from  $\mathbf{CM}^\otimes$ .

**Principle:** “I am not just working with plain  $\infty$ -categories; I have a class of ‘special’ edges (inert maps) that must be preserved. The model structure is designed so that these edges behave like cofibrations or fibrations depending on context.”

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## The Overarching Principle

In homotopy theory (and specifically Lurie's approach), **fibrations are tools to solve lifting problems**.

You choose the fibration type according to:

### 1. What kind of lifting do I need?

- Lifting inner horns  $\rightarrow$  inner fibration.
- Lifting all horns  $\rightarrow$  Kan fibration.
- Lifting edges in a controlled way (cocartesian)  $\rightarrow$  cocartesian fibration.
- Lifting everything (including boundaries)  $\rightarrow$  trivial fibration.

### 2. What structure am I preserving?

- Just composition  $\rightarrow$  inner fibration.
- A functorial family  $\rightarrow$  cocartesian/cartesian.
- A weak equivalence  $\rightarrow$  trivial fibration.
- Operadic inert edges  $\rightarrow$  marked fibration/cofibration.

### 3. What is the base category?

- If base is  $F_*$  (finite pointed sets) and you are doing operads, you almost always use an inner fibration with marked inert edges (the  $\infty$ -operad structure).
- If base is an arbitrary  $\infty$ -category and you want a “family of  $\infty$ -categories” over it, you use a cocartesian fibration.

**Practical rule of thumb:**

- **See an inner fibration**  $\rightarrow$  “I’ m just moving between  $\infty$ -categories, nothing fancy.”
- **See a cocartesian fibration**  $\rightarrow$  “I’ m doing a Grothendieck construction; objects in the fiber can be pushed forward.”
- **See a trivial fibration**  $\rightarrow$  “This map is an equivalence, and it’ s nice enough to lift any diagram.”
- **See a marked fibration / trivial cofibration**  $\rightarrow$  “I’ m in operad world; inert edges matter.”

Once you internalize this, you can read Glasman’ s paper (or any part of Lurie’ s *Higher Algebra*) by focusing on **what lifting problem is being solved** – the fibration name tells you what solutions exist.

用户提问

make it a simple principal for helping me to master fibrations

DEEPSEEK 回答

Here’ s a **single master principle** to keep in your pocket. Once you grasp it, all fibrations become predictable.

The Master Principle

A fibration is defined by *which simplices (or edges) you are guaranteed to be able to lift uniquely (up to homotopy)*. Its name tells you exactly that lifting class.

So when you read “inner fibration” , “cocartesian fibration” , “trivial fibration” , etc., always ask: “What am I allowed to lift?”

Simple decision tree (for 90% of cases)

| If you need to lift...                                          | Then you need a...    | Intuition                                      |
|-----------------------------------------------------------------|-----------------------|------------------------------------------------|
| Inner horns $\Lambda^n_k$ ( $0 < k < n$ )                       | Inner fibration       | Basic functor / $\infty$ -category             |
| Inner horns + certain edges (cocartesian)                       | Cocartesian fibration | A functor to $\text{Cat}_\infty$ (pushforward) |
| All horns (including outer)                                     | Kan fibration         | Fibration between $\infty$ -groupoids          |
| All boundary inclusions $\partial\Delta^n \rightarrow \Delta^n$ | Trivial fibration     | A weak equivalence that lifts everything       |
| Marked anodyne extensions (special inert edges)                 | Marked fibration /    | Operad / preoperad                             |



| If you need to lift... | Then you need a... | Intuition |
|------------------------|--------------------|-----------|
|                        | <b>cofibration</b> | maps      |

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## Applied to Glasman' s paper

- **Lemma 4** needs an **inner fibration** + cocartesian edges → uses **cocartesian fibration** (to lift a whole cone).
  - **Proposition 3** needs to extend maps → uses **trivial cofibration** in the **marked** model structure (to guarantee extension without changing homotopy type).
  - **Proposition 7** needs restriction to be an equivalence → uses **trivial Kan fibration** (lifts all boundaries → equivalence of mapping spaces).
- 

## The ultimate shortcut

**Don' t memorise definitions – memorise lifting problems. Every fibration is a solution to a specific “can I fill this shape?” question.**

- Inner → “Can I compose?”
- Cocartesian → “Can I push objects forward along a map?”
- Trivial → “Can I uniquely lift any diagram up to homotopy?”
- Marked → “Can I preserve special edges (like inert maps)?”

Once you know **what** you want to lift, the fibration name is automatic. That' s the principle.